



East West University (EWU)
Dept. of Mathematical and Physical Sciences (MPS)
Course Title: Coordinate Geometry and Vector Analysis
Course Code: MAT 104
Semester: Fall-2025, Section: 14
Assignment

Date of Submission: 04/12/2025

1. A directional antenna transmits in a direction $\vec{d} = (3,1,2)$. A signal from a user arrives as a vector $\vec{s} = (4,2,1)$.
 - (a) Find the projection of $\vec{s}$ on $\vec{d}$.
 - (b) Explain how this projection helps in beamforming for 5G networks.
2. A mobile robot moves in a terrain represented by a potential field
$$f(x,y) = 4x^2 + y^2 - 10x.$$
 - (a) Find the gradient vector ∇f .
 - (b) Explain how the robot uses $-\nabla f$ for gradient descent navigation to avoid obstacles.

*****Good Luck*****